

Deep Convolutional Networks for Skin Lesion Classification

(2023)

Problem

Classifying dermoscopic images of skin lesions into benign and malignant categories.

Method

A CNN-based pipeline trained end-to-end on dermoscopic images.

Dataset

ISIC dataset (International Skin Imaging Collaboration).

Evaluation

Accuracy, sensitivity, and AUC on a held-out test split.

Results

The proposed CNN achieved 91.2% accuracy and 0.94 AUC, outperforming a Random Forest baseline built on handcrafted texture features.

Limitations

Performance drops noticeably on darker Fitzpatrick skin tones due to dataset imbalance, an open problem for future work.